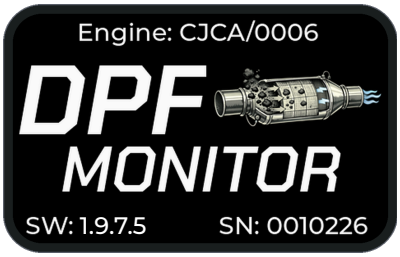
User Manual

DPF Load Monitor Mini

for VAG group TDI common-rail engines (UDS diagnostics)



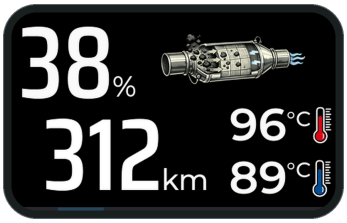
1 About the device

DPF Load Monitor Mini is a compact 1.64" AMOLED touch display that plugs into the OBD-II socket and reads live data straight from the engine ECU: DPF load, soot mass, DPF temperatures, differential pressure, distance and time since the last regeneration, oil and coolant temperature, boost, fuel pressure and more. It recognises the engine automatically, switches on with the engine and off when the engine stops, signals DPF regenerations, keeps a regeneration history and reads / clears fault codes.

2 Installation and first start

1. **Start the engine first**, then plug the device into the vehicle's **OBD-II diagnostic socket** and route the cable so it does not obstruct the driver.
2. With the engine running the welcome screen appears (detected engine code, software version **SW**, serial number **SN**), then the main screen with live values.
3. Set the clock: Menu → **Set Clock**. If you do not want to use the clock, you do not need to set it.
4. When the engine is switched off the device shows *bye bye*, the screen goes dark and the device goes to sleep. It wakes up by itself at the next engine start.

⚠ **Important:** make the first connection to the OBD-II socket **with the engine running**. With the engine off the vehicle CAN bus is asleep and the device will not start.



Main screen



Engine stopped - going to sleep

3 Touch gestures

Swipe left / right	Main screen: change page (7 pages). Advanced: change page. History: scroll.
Swipe down	Main screen: open the **Menu**.
Swipe up	Menu: back to the main screen. Page 6: current → MAX → MIN. Page 7: reset MAX.
Swipe right	Any screen opened from the Menu: back to the Menu (graphs: back to the graph list).
Tap	Main screen: toggle **Day / Night** brightness (when Auto brightness is off). Buttons: open / confirm.
Hold 1 s	Main screen: **Standard / Big** layout. Menu buttons: extra functions (see Menu).

4 Automatic operation

The device switches on by itself after the engine is started, reads the engine type and connects to the ECU. After the engine is switched off it shows the trip statistics (if enabled), then *bye bye*, the screen goes dark and the device goes to sleep. Min/max values are counted again from each engine start.

5 Screen map



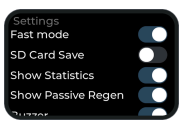
Main screen
7 pages

swipe down ↓ swipe up ↑



Menu

→



Settings



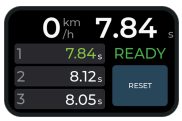
Set Clock



History



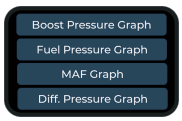
DTC



0-100



Advanced



Advanced (hold)



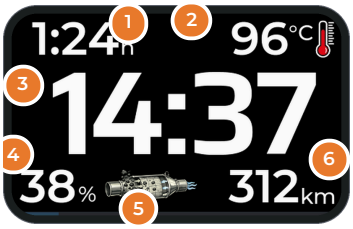
Info

↩ swipe right = back to the Menu

6 Main screen - Standard layout

Swipe left/right to move between the seven pages. After start-up the device shows page 2. Example values are shown.

Page 1 - Clock



1. trip time since engine start, h:mm
2. engine oil temperature, °C
3. clock
4. DPF load, %
5. DPF state icon (white / **orange** / **red**)
6. distance since last regeneration, km

Page 2 - Overview



1. DPF load, %
2. DPF state icon
3. distance since last regeneration, km
4. oil temperature, °C
5. coolant temperature, °C

Page 3 - Soot



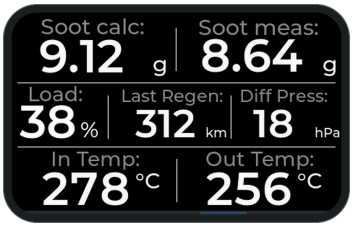
- DPF load %, distance since regeneration km
- **Calc / Meas** - calculated / measured soot mass, g
- **mg/km** - soot build-up per kilometre
- ΔP - DPF differential pressure, hPa

Page 4 - DPF temperatures



- DPF load %, distance since regeneration km
- DPF inlet / outlet temperature, °C
- ΔP - differential pressure, hPa

Page 5 - Regeneration page



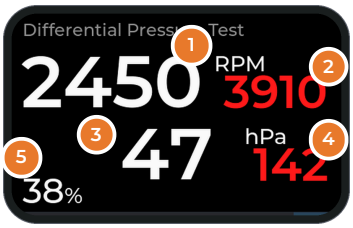
- soot calc / meas g, load %, last regen km, ΔP hPa, DPF inlet / outlet temperature °C
- shown automatically when a regeneration starts (if *Switch page to regen* is on)

Page 6 - All values



- DPF in / out, oil, coolant, intake temperatures; km and min since regeneration; soot calc / meas; ΔP; oil ash; battery voltage
- swipe up: MAX → MIN → current

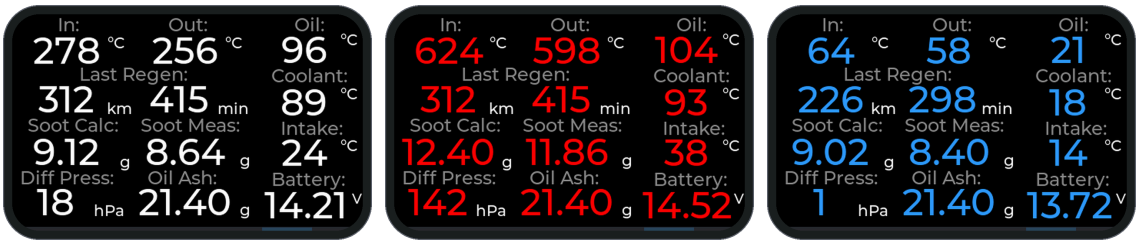
Page 7 - Differential pressure test



1. current engine speed, RPM
2. RPM at which the maximum ΔP was recorded
3. current ΔP, hPa
4. maximum ΔP, hPa (swipe up = reset)
5. DPF load, %

7 Page 6 - current / MAX / MIN view

Swipe **up** on page 6 to cycle all values: **current** (white) → MAX (red) → MIN (blue) → current. Extremes are counted from engine start. Changing the page returns to current values.



current MAX MIN

8 Big layout

Hold your finger on the main screen for 1 s to switch to the Big layout - the same seven pages with larger digits and short captions. Hold again to go back. The choice is remembered.



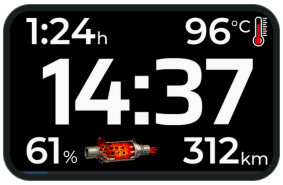
1 2 3 4 5 6 7

TRIP = trip time · OIL / COOLANT / INTAKE = temperatures °C · LOAD = DPF load % · REGEN = km / min since regeneration · SOOT CALC / MEAS = soot mass g · SOOT RATE = mg/km · IN / OUT = DPF inlet / outlet °C · dP = differential pressure hPa · OIL ASH = oil ash · BATTERY = V · MAX = maximum

9 DPF regeneration

When the engine ECU starts a DPF regeneration, the device gives **two beeps**, the **DPF icon turns red** (pages 1-2) and these values turn **red**: DPF load on pages 3 and 7, DPF inlet / outlet temperatures on pages 4, 5 and 6 (also in the Big layout). With *Switch page to regen* on, the display jumps to the regeneration page. When the regeneration is finished the device gives **one beep** and the colours return to normal; the regeneration is saved in the History.

Passive regeneration (with *Show Passive Regen* on): when soot burns off by itself at high exhaust temperatures, the icon and the same values are shown in **orange**.



Page 1



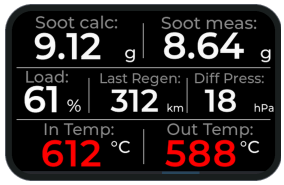
Page 2



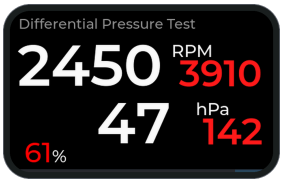
Page 3



Page 4



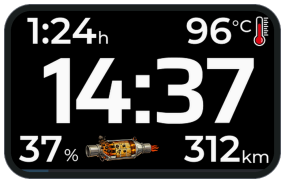
Page 5



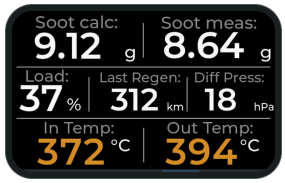
Page 7



Big - page 5



Passive - page 1



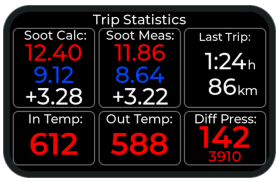
Passive - page 5

10 Menu



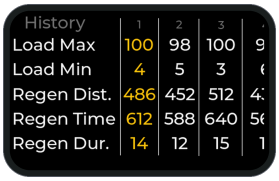
- **Settings**, **Set Clock**, **History**, **DTC**, **0-100**, **Info** - tap to open.
- **Advanced** - tap: extra engine data; **hold 1 s**: graphs.
- **History** - **hold 1 s** to delete the whole history.
- **Update** - **hold 1 s** to restart in firmware-update mode (screen goes dark). Use only when installing new firmware according to its instructions.

11 Trip statistics



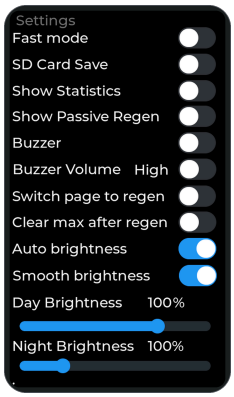
Shown after the engine is switched off (if *Show Statistics* is on): calculated and measured soot - **max**, **min** and change during the trip; trip time and distance; maximum DPF inlet / outlet temperature and maximum ΔP with the RPM at which it occurred.

12 History



The last 10 regenerations; column **1** (yellow) is the newest. **Load Max / Min** - DPF load at the start / end of the regeneration, %. **Regen Dist.** - km since the previous regeneration. **Regen Time** - minutes since the previous regeneration. **Regen Dur.** - duration of the regeneration, min. Swipe left/right to see older entries.

13 Settings



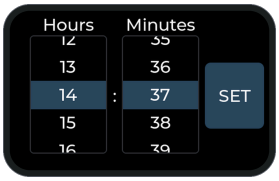
- **Fast mode** - faster refresh of the DPF values.
- **SD Card Save** - records the data to a microSD card once per second (CSV files, a new file after each regeneration). This rate is intended for recording the whole DPF filling cycle. Available only with a card inserted; a blinking **green mark** at the top edge shows that data is being saved.
- **Show Statistics** - trip statistics after the engine is switched off.
- **Show Passive Regen** - shows passive regeneration in orange.
- **Buzzer** / **Buzzer Volume** - sound signals on/off, volume High / Low.
- **Switch page to regen** - jump to the regeneration page when a regeneration starts.
- **Clear max after regen** - reset the MAX values of the ΔP test after each regeneration.
- **Auto brightness** - the light sensor (on the top of the housing) switches Day / Night brightness.
- **Smooth brightness** - smooth transition between Day and Night brightness instead of a step change. Active only with Auto brightness on.
- **Day / Night Brightness** - screen brightness levels.
- Settings are saved automatically. Scroll the screen to see all items.

14 DTC - fault codes



Fault codes of the engine ECU are read automatically when the screen opens (up to 10, scroll to see more). **Read** - read again. **Clear** - delete the fault codes in the engine ECU. The status (e.g. *No DTC*, *DTC Cleared*) is shown at the top right.

15 Set Clock



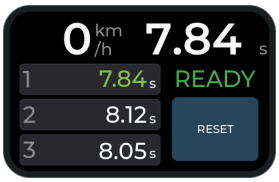
Set hours and minutes with the rollers and tap **SET**. The device has no built-in battery, so the time is not kept after it is unplugged - set it again after each disconnection.

16 Info



Device name, hardware / software version, serial number, detected engine code and ECU part number / software. The QR code opens **dpfloadmonitor.pl**.

17 0-100 km/h

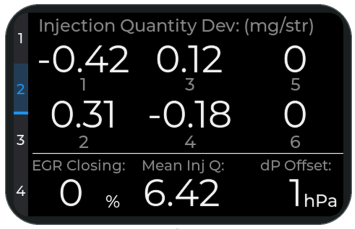


1. Stand still - status **READY**.
2. Accelerate - the time is measured; at 100 km/h a beep sounds and the result is saved.
3. The last 3 results are listed, the best one in green. Tap **RESET** (standing still) for a new run, **hold RESET** to clear the results.

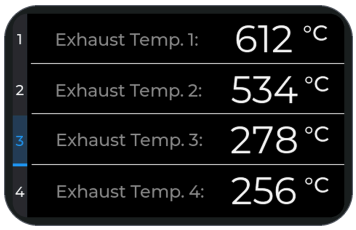
18 Advanced



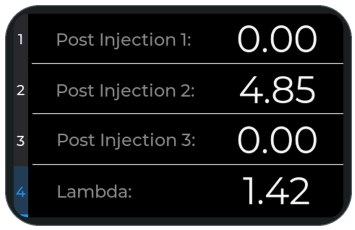
1



2



3

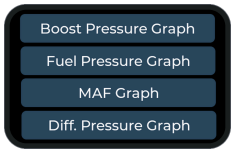


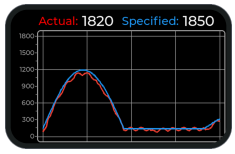
4

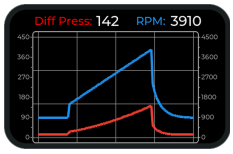
- **Page 1** - boost pressure (mbar, with bar graph), fuel pressure (bar), air mass MAF (mg/str): actual and specified.
- **Page 2** - injection quantity deviation of cylinders 1-6 (mg/str), EGR closing (%), mean injection quantity **Mean Inj Q** (mg/str), ΔP sensor offset (hPa).
- **Page 3** - exhaust temperatures 1-4 (°C).
- **Page 4** - **Post Injection 1 / 2 / 3**: post injection quantity (mg/str) or control duration when the title shows **[ms]** (engine dependent), and **Lambda** - from the oxygen sensor or calculated by the ECU. During an active DPF regeneration post injection 2 or 3 rises.

A parameter the engine ECU does not provide is greyed out and shows 0.

19 Graphs

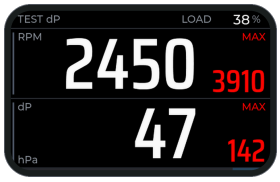






Hold **Advanced** for 1 s and choose a graph. Boost, fuel pressure and MAF: **actual** vs **specified**. Diff. pressure graph: **ΔP** (left scale) and **RPM** (right scale) - handy during the ΔP test.

20 Differential pressure (ΔP) test



1. Warm the engine up fully and go to page 7.
2. Swipe up to reset the MAX values.
3. Only where it is safe and legal: in 3rd gear accelerate at full load up to about 4000 RPM.

⚠ Do not operate the device while driving. Perform the ΔP and 0-100 tests only on a closed or empty road, observing traffic rules.

21 microSD card - data logging



When **SD Card Save** is enabled in **Settings**, the device appends one row of data to a CSV file on the microSD card every **1 s** (data is physically written to the card every 5 s). A blinking **green dot** at the top edge of the screen means logging is active.

Card: microSD formatted as **FAT32** or **exFAT**. The card can be inserted and removed while the device is connected - the photo shows which way to insert it. One hour of driving is about 250 kB of data, so even a small card lasts a very long time.

File names: dpf_log_0.csv, dpf_log_1.csv, ... - each new file gets a number one higher than the highest one on the card (up to dpf_log_999).

When a new file is created

- after the **card is inserted**,
- after the **device starts / restarts** (e.g. it is plugged into the OBD-II port again),
- after **every DPF regeneration ends** - so one file covers one filter loading cycle, from regeneration to regeneration.

After switching the engine off and on again, or turning **SD Card Save** off and on, logging **continues in the same file**.

Removing the card

It is safest to remove the card with the engine off (device asleep, file closed) or after turning **SD Card Save** off. Removing it while logging may lose the last few seconds of data.

Write errors

If a write error occurs, the **SD Card Save** switch turns grey and logging is disabled. To restore it, reset the device - unplug it from the OBD-II port and plug it in again.

CSV file columns

DPF_in	DPF inlet temperature, °C
DPF_out	DPF outlet temperature, °C
Diff_press	differential pressure ΔP, hPa
Last_regen_km	distance since last regeneration, km
Last_regen_min	time since last regeneration, min
Oil_temp	oil temperature, °C
Soot_calc	calculated soot mass, g
Soot_meas	measured soot mass, g
Coolant	coolant temperature, °C
Batt	battery voltage, V
Engine_time	engine run time since start, min
Intake	intake air temperature, °C
Engine_speed	engine speed, RPM
Vehicle_speed	vehicle speed, km/h
Post_Inj2	post injection 2 (raw ECU value)
Post_Inj3	post injection 3 (raw ECU value)
Oil_Ash	oil ash, g or l (engine dependent)

Parameters not supported by a given engine are logged as 0.

Viewing the logs

The file is plain CSV: the first row holds the column names, the separator is a comma, the decimal separator is a dot, **1 row = 1 second**. The most convenient way to look at it is as graphs in an **online log viewer**, e.g. datazap.me or csvplot.com - just load the file and pick the parameters. Offline: MegaLogViewer, Excel or LibreOffice (if your Excel uses a decimal comma, open the file via **Data → From Text/CSV** and set the comma as separator and the dot as decimal separator).

Worth comparing: growth of Soot_calc and Soot_meas, Diff_press vs Engine_speed, DPF_in / DPF_out temperatures during regeneration. Post_Inj2 / Post_Inj3 above 0 means active post injection (regeneration).